

Literature Review on LLM Prompting and Societal Implications

Societal aspects and potential impacts, of AI in training, inference, and prompting of Large Language Models

This study group exercise is a part of the course LLMs for Developers (*Societal Aspects*). The exercise focuses on how interaction with LLMs through prompting can influence the quality, reliability, and societal impact of AI-generated information. The project examines three challenges: **ambiguity and semantic drift, response hallucination, and prompt injection**.

Background

LLMs increasingly influence communication, learning, software development, and knowledge discovery. This project examines how ambiguity, hallucination, and prompt injection affect the trustworthiness of AI systems and their impact on individuals, organisations, and society.

AMBIGUITY/SEMANTIC DRIFT	HALLUCINATION	PROMPT INJECTION
<i>Can an unclear AI response lead to the wrong conclusion? Or progressively renewed conclusions?</i>	<i>Can a false AI answer become accepted as truth?</i>	<i>Can AI systems be compromised and manipulated through prompts?</i>
Explore how multiple interpretations affect decisions and communication.	Investigate fabricated facts, detection methods, and misinformation risks.	Examine attack patterns/techniques, vulnerabilities, and security implications.

A Societal Risk Pathway or Derailment?

Ambiguous prompts → Ambiguous responses → User misinterpretation → Hallucinated or incorrect beliefs → Manipulation opportunities → Societal impact →→ Ambiguous prompts → Ambiguous responses → ...

Learning Objectives

Search and evaluate from literature:

- Explain the concepts of ambiguity, semantic drift, hallucination and prompt injection.
- Synthesize evidence from literature, in the context of training and inference (prompting) of LLMs.
- Discuss potential implications of these potential prompting artifacts.
- Communicate your findings to peers: assume something like a company setting where your group is in the IT department, reporting to the CEO, CSO and CHRO. Reporting is to the class-room, of course.

Deliverables

- One Power Point presentation (per group) -- max. 6 slides, max 10 minutes.
- One synopsis (per group) -- max 6 pages + coverpage.

Per-group hand-in: Thursday evening of week 5, September 23rd, [by email to jatr@ek.dk](mailto:jatr@ek.dk).

NB: You may use AI in your work, cautiously. You must be able to account in dialogue for your deliberations.

Literature Search

Search for example Google Scholar, ACM Digital Library, arXiv. Prioritize articles, conference proceedings, blogs, and reviews, preferably from the last 5–7 years. Some inspiration on ItsLearning: [articles and links](#).

Scope

Groups of 4–6 students. Workload: approximately 0.5 FTE week per group (this is about 23 hrs per group -- or 4.5 hours for a group of five). Focus on analysis and synthesis of existing research, not experiments.